

CIRQL — Build Plan

"An ocean of circles that never ends — you light it with real taps, glowing Wonders, and the friends in your Cirql."

CIRQL is the flagship: a persistent, magical, social world of concentric island-rings. You start at **The Hearth** and sail outward forever. This plan is the working guide — checkoff lists mirror the Linear project "**CIRQL — Flagship World**" (team CHR). **Status (2026-07-07): M0–M10 COMPLETE & LIVE + M11 underway**. The playable slice grew into a real multiplayer world — live presence + chat, co-op campaigns/party-finder, endless procedural rings + generated quests + daily/seasonal loops + a Myst puzzle, themed seasonal islands with purposeful landscape, and interactive voyages into caves/treetops/clouds. Everything is live at [/cirql](#) on [chr-5-7-phase0-run-locally](#) (auto-deploys to Render). Currency shows as **SPARQ/SPARQS**.

Progress log

- **M1 · The World & Movement — DONE & LIVE** (commit [b664883](#)). Walkable Hearth island: cottage, Wonders monolith, Ferra (NPC), Cirql lantern ring, movement, minimap, glow, interact → dialog/toast.
- **M2 · Identity & Persistence — DONE & LIVE** (commit [b09c79c](#)). Save/resume via the [cirql](#) game_progress key; first-run character creator (avatar moved into CIRQL) + "Look" re-edit; expanded avatar (hat styles, aura glow, companions, bigger palettes).
- **M3 · Quest System (backbone) — DONE & LIVE** (commit [32e4d9e](#)). Reusable registry (defs in code, progress in the [cirql](#) blob — no migration), objective model + chain, reward→sparks hook, objective tracker HUD, waypoint chevron + minimap dot, NPC give-via-dialog, quest-log panel.
- **Feel/UX polish — DONE & LIVE**: smooth-text overlay (readable UI on phones, [e3905ee](#)), full-bleed responsive viewport + 4-direction facing ([d9e2b27](#)), full-screen immersive layout + tap-absorbing control tray ([a2e69e4](#)), and the **full-screen sea chart** (tap the minimap; [3dd6f46](#), CHR-250).
- **M4 · First Quest Chain (onboarding) — DONE & LIVE** (commit [677bf40](#)). Find Your Feet → The Lantern Path → The CirqlCade Door, auto-chaining; first-run auto-accepts quest 1.
- **Brand**: flagship wordmark **CIRQLVERSE** (CIRQL white + VERSE half-size violet); the in-world arcade building renamed **CirqlCade** with a cosmic mystery-dome roof.
- **M5 · CirqlCade (in-world arcade) — DONE & LIVE** (commit [26eead2](#)). Enter CirqlCade → a hall of all ~57 live cabinets (from the registry); tap one → it plays **embedded in an iframe over** [/cirql](#) (never leaves CIRQLVERSE); "← CirqlCade" exits to the hall and the game's own /arcade back-link is caught + routed back; **a spark a play** on exit. Verified end-to-end (0 errors, tsc clean).
- **M6 · Sparks Economy & Your Cirql — DONE & LIVE** (commits [9f4c0e4](#), [df76f4a](#)). World Energy meter, your-Cirql lanterns + Cirql panel (Invite + "share a light"), capped spark-a-play, and **spend-sparks cosmetics** (buy-in-place in the creator; owned persists). **Only CHR-245 (retire the platform-wide arcade reward system) remains — intentionally deferred** to its own careful pass (big blast radius).
- **M7 · Playable Slice Hardening — DONE & LIVE** (commit [6b52ba9](#)). Full-loop smoke (arcade banner → world → create → quests → CirqlCade → play → sparks → spend → resume) verified; tsc clean + mobile pass; **CIRQLVERSE surfaced via a flagship banner on** [/arcade](#); beta-test checklist doc created.
- 🎉 **THE PLAYABLE SLICE (M0–M7) COMPLETE & LIVE**. Then the deeper roadmap shipped:
- **M8 · Social — presence + chat — CORE DONE & LIVE** (commit [d23d975](#)). New [server/cirql-presence.ts](#) = a ring-scoped live room at [/ws/cirql](#) (grown from [/ws/town](#)): real travellers walk the Hearth together (avatars, name tags, chat bubbles, smooth interpolation), a header chat bar, live online count, and **"share a light"** (walk up + E → both Hearths light a lantern — the locked social payoff). *DM channel + full channel-switcher/moderation (CHR-248/249) are deferred; global + party chat ship.*
- **M9 · Campaigns & Party Finder — DONE & LIVE** (commit [a5ee3b3](#), CHR-251/252/253/254). Co-op **campaigns** ([cirql-campaigns.ts](#)), a **Campaign Board** with **structured, PII-free** matchmaking posts (pick a campaign + preset tags + new-player-friendly — no freeform, kid-safe), host/seeker directions, ask→accept + invite→accept handshakes, **parties** with a shared teal waypoint + synced step progress + a members-only **party chat channel** + rewards; leave/disband/late-join; rate limits + minimal block/report.
- **M10 · Endless content engines — DONE & LIVE** (commits [bff0024](#) → [89cfbc9](#), CHR-255/256/257/258). **Procedural rings** ([cirql-ring-gen.ts](#): sail outward forever — seeded biomes/props/docks, growing radius, arrival cinematic, resume); **quest-template generator** (each ring's keeper offers a deterministic kindle/discover/wayfind quest via a dynamic registry); **daily + weekend loops** ([cirql-daily.ts](#): rotating daily task, streak, a weekend Lantern-Festival ×2); a **Myst rune puzzle** on the Hearth (read the runestone clue → light the runes → the sealed shrine opens).
- **World polish (owner feedback)**: themed **seasonal islands** (meadow/tropical/winter/desert/autumn/woodland/coast/ember — biome stride so every shore differs), **world-space ambient critters** with per-creature motion (butterflies/bees/dragonflies/grasshopper-hops/fireflies/gulls/snow/embers), **purposeful landscape** (fence corrals + rows, flower beds, tree groves, rock cairns, paths), **solid collision** (walk around trees/rocks/ponds/fences), **cross-ring quests + campaigns**

(objectives/steps span rings; off-ring the waypoint points to the dock), **repeatable quests/campaigns** (redo for ~¼ reward), and a robust **lantern-quest fix** (per-quest `litForQuest`).

- **M11 · Depth & Delight — STARTED: CHR-265 Interactive voyages — DONE & LIVE** (commit `be24ce3`): a `portal` prop (cave/hollow-tree/cloud-stair) travels to a **sub-map** (The Undervault cavern / The High Canopy / The Cloud Reach) and back — reusing the sailing primitive (a portal is a dock to an off-index ring). Emotes (CHR-260, roster ready) + cutscenes (CHR-264) + more wishlist remain.
- **Currency rename:** flagship display is now **SPARQ/SPARQS** (code identifiers stay `sparks`).

Locked decisions (owner, 2026-07-06 — "ALL IN")

- **The Wonders** = the 50 arcade cabinets folded IN-WORLD as enchanted monoliths you *attune* to (press E → the cabinet opens as an in-world overlay; you never leave CIRQL). Found on the first quest, which is the movement tutorial.
- **One currency: SPARKS.** All rewards come out of the arcade's own system and unify under sparks. **Earn** ~a spark a play + a small capped performance bonus (and, Phase 2, real-world taps). **Spend** in an in-CIRQL reward catalog (avatar cosmetics, Hearth décor, perks). Sparks are the games *and* in-platform reward.
- **Avatar lives in CIRQL.** The chooser moves out of the arcade into a **first-run character-creation** step (sign-up / first play), with **lots more fun choices** — auras/glows, patterns, hats/crowns, companions — despite the circular art. Free at creation + spark-unlockable extras.
- **Your Cirql** = your friends, shown as **lanterns lighting your Hearth** (dim → beacon). Grown by **meeting people in-world** ("share a light", a chat-free beacon) **and inviting real friends** (rewarded-referral, reskinned as light).
- **Chat is channelled** so the feed never jams: **Global** (ambient) · **Party/Group** (campaigns & friends) · **Direct Messages**. Needs the live presence layer, so it becomes milestone **M8**.
- **Look:** supersampled 16-bit RetroEngine + the neon glow-up kit — "modern-but-16-bit". Approved 5-frame lookbook + playable in-game mockup.

Completed — reusable foundations (M0)

- ✓ `/town` websocket prototype — presence + move + chat proven (CHR-211)
- ✓ RetroEngine, supersampled (SS=1.5) (CHR-212)
- ✓ 50 arcade cabinets + registry + arcade-shell — become the Wonders (CHR-213)
- ✓ Avatars + shared joystick + `game_progress` save table (CHR-214)

M1 · The World & Movement — DONE

- ✓ `/cirql` route + lazy page scaffold (CHR-215)
- ✓ World engine: canvas render loop + camera (CHR-216)
- ✓ Concentric-ring data model + ring configs (CHR-217)
- ✓ The Hearth (ring 0) authored map + props (CHR-218)
- ✓ Player movement: WASD + click + joystick + edge collision (CHR-219)
- ✓ Minimap: concentric rings + fog + player dot (CHR-220)
- ✓ Ambient neon polish (glow-up kit) (CHR-221)
- ✓ Full-screen sea chart (tap minimap to expand) (CHR-250)

M2 · Identity & Persistence — DONE

- ✓ Cirql state model + save/load endpoints (`cirql` key) (CHR-222)
- ✓ Load-on-enter + autosave + resume (CHR-223)
- ✓ Character creation on first entry — move avatar chooser into CIRQL (CHR-242)
- ✓ Expand avatar customization — more choices, fun, circle-friendly (CHR-243)

M3 · Quest System (reusable backbone) — DONE

- ✓ Schema/persistence: quest progress in the `cirql` state blob (no migration) (CHR-224)
- ✓ Quest registry + progress engine + reward→sparks hook (CHR-225)
- ✓ Client: quest log + objective tracker HUD (CHR-226)
- ✓ Client: NPC quest-givers + waypoint markers (CHR-227)

M4 · First Quest Chain (onboarding = movement tutorial) — DONE

- ✓ Quest 1 — "Find Your Feet" (teaches movement) (CHR-228)
- ✓ Quest 2 — "The Lantern Path" (teaches interact) (CHR-229)
- ✓ Quest 3 — "The Wonders Door" (find + enter the arcade) (CHR-230)
- ✓ First-run onboarding scripting hook (CHR-231)

M5 · CirqlCade (in-world arcade) — DONE

- ✓ CirqlCade hall — all live cabinets from the registry (CHR-232)
- ✓ Press-E launches a cabinet embedded over `/cirql` (iframe) (CHR-233)
- ✓ Return-to-world flow (catches the game's `/arcade` back) (CHR-234)
- ✓ Score → sparks (spark-a-play MVP; full normalization in M6) (CHR-235)

M6 · Sparks Economy & Your Cirql — CORE DONE

- ✓ Sparks currency + World Energy meter (CHR-236)
- ✓ Your Cirql: friends as lanterns around the Hearth (CHR-237)
- ✓ Invite-a-friend hook + "share a light" beacon stub (CHR-238)
- ✓ Earn sparks from play (spark-a-play + daily-capped) (CHR-244)
- ✓ Spend sparks: in-CIRQL cosmetics (buy-in-place in the creator) (CHR-246)
- Unify rewards → sparks (retire the arcade's separate reward system) (CHR-245) — deferred (platform-wide; own pass)

M7 · Playable Slice Hardening — DONE

- ✓ Full-loop smoke test (CHR-239)
- ✓ tsc clean + mobile pass + perf (CHR-240)
- ✓ Beta-test checklist + deploy + surface `/cirql` in nav (CHR-241)

M8 · Social — presence & chat channels — CORE DONE

- ✓ Live presence in-world (`/ws/cirql` , room per ring) (CHR-247)
- Chat channel model: Global / Party / DM (CHR-248) — **🟢 global + party chat done; DM deferred**
- Channel switcher UI + safety/moderation (CHR-249) — **🟢 Global/Party toggle + structured tags + rate limits + minimal block shipped; full switcher/DM/profanity-filter/persistent report-block deferred**

M9 · Campaigns & Party Finder (co-op) — DONE

- ✓ Campaign model: co-op multi-step quests (party-scoped) (CHR-251)
- ✓ Campaign Board: post / browse / join (+ new-player-friendly) (CHR-252)
- ✓ Party formation + shared waypoint + shared progress (CHR-253)
- ✓ Matchmaking requests + safety (tags, report/block) (CHR-254)

M10 · Endless content engines — DONE

- ✓ Procedural rings (infinite outward biomes) (CHR-255)
- ✓ Quest-template generator (infinite authored-feeling quests) (CHR-256)
- ✓ Daily + seasonal loops (rotating dailies, events, drops) (CHR-257)
- ✓ Myst-style puzzle quests / campaigns (CHR-258)

M11 · Depth & Delight (wishlist — owner-approved) — STARTED

Positioning decision: **Balanced** (family-friendly / chat-free-first defaults + cosmetic-only + kindness mechanics, with fuller chat + competitive edges for older players).

- ✓ **Interactive voyages: tunnels/underground, treetops & clouds** (CHR-265) — DONE & LIVE
- Hearth décor + visiting friends' Hearths (CHR-259)
- Emotes + chat-free expression (CHR-260) — full roster drafted, not built

- Tapped businesses become in-world landmarks (loyalty ↔ world) (CHR-261)
- Sailing as a mini-journey (interactive voyages) (CHR-262)
- Jump / hop movement (traversal + expression) (CHR-263)
- More mini-journeys & cutscenes (expand) (CHR-264) — brainstorm captured

Architecture notes (as built through M11)

- `client/src/game/cirql-world.ts` — declarative rings/props + `PropTypes`. **The authored Hearth (ring 0) + every prop type live here.** Prop types now include `hearth/wonders/npc/tree/lantern/crystal/dock/marker/tablet/rune/shrine/gathering/theater/rock/pond/flower/fence/path/portal`.
- `client/src/game/cirql-ring-gen.ts` — **procedural rings + sub-maps** (CHR-255/265). `generateRing(index)` seeds `biome/palette/props/docks`; radius grows with index; landscape uses **formations** (`placeGrove/placeFlowerBed/placeRockCairn/placeCorral/placeFenceRow`). **Sub-maps:** `getRing(i)` returns the Hearth for 0, a surface ring below `SUB_BASE`, or a **sub-map** (`generateSubMap` `cave/tree/cloud`) for big indices; `isSubMap/parentOf/subKindOf/subIndex` encode `kind*offset + parentIndex`. A **portal is a dock** to an off-index ring.
- `client/src/game/cirql-quests.ts` + `cirql-quest-gen.ts` — quest registry (authored chain + `registerQuest` / `allQuests` **dynamic** ring quests). Objectives carry an optional `ring` (cross-ring) + kinds `reach/interact/enterWonders/lightLanterns/solvePuzzle`. Repeatable via `doneOnce`; `lightLanterns` uses a **per-quest** `litForQuest` set (never blocked by stale global lit).
- `client/src/game/cirql-campaigns.ts` — co-op campaigns; steps carry `ring` + `at` (a prop id resolved on that ring) for **cross-ring** party voyages.
- `client/src/game/cirql-daily.ts` — daily task + seasonal/weekend event (deterministic per UTC day).
- `client/src/game/cirql-theater.ts` — the Cirql Drive-In's fake movie posters.
- `client/src/game/cirql-world-engine.ts` — `CirqlWorldEngine` extends `RetroEngine`. Hooks: `onInteract`, `onLocalMove` (autosave), `onSail(ring,maxRing)`, `onQuestComplete(q,firstTime)`, `onQuestChange`, `onPresence` / `onShareLight` (multiplayer), `onPartyArrive`; `setStats` / `setPartyTarget` / `setPartyMembers` / `addRemote...` / `toast`. Interact key = **E**. In DEV the engine is on `window.__cirql` (TS `private` fields are runtime-readable — used for smoke tests).
- `server/cirql-presence.ts` — the `/ws/cirql` room: ring-scoped presence + move + chat (global/party channels) + share-a-light + the M9 **party/campaign-board state machine** (in-memory, no DB). Wired into the `noServer` upgrade router in `server/routes.ts` beside `/ws/town` + `/ws/mp`.

Gotchas to respect

- **DB migrations:** `db:migrate` is broken on the shared Neon DB; `db:push` truncates. All CIRQL state rides in the `game_progress` `cirql` blob — **no new tables**; the whole multiplayer/party layer is in-memory.
- **Persist everything through the blob:** `buildState` / `applyState` in `cirql.tsx` must list each field (`ring/maxRing/x/y/quests/lit/litForQuest/doneOnce/doneCampaigns/...`) — a field left out silently isn't saved.
- **Lantern quests:** never gate quest-lantern lighting on the shared persistent `lit` set — use the per-quest `litForQuest` (reset on accept). (This was the "3 lights on, can't light another" bug.)
- **Dev server:** `tsx` does not hot-reload server code — restart after server edits (kill the port-5000 PID). Client hot-reloads via Vite. **Rapid full page reloads trip the express rate-limiter** (`max 1000/15min`, `server/index.ts`) → 429s/blank page → restart the dev server to reset; prefer HMR while testing.
- `apiRequest` **returns a raw** `Response` — the cirql page uses raw `fetch`.
- **Don't shadow RetroEngine base methods/fields** (`px`, `ring`, `last`) — world fields are `posX/posY/curRing`.
- **Test-account state:** a saved `seenIntro:false` shows the first-run creator (gates presence join) — set it true in the `cirql` save to smoke multiplayer.
- **Deploy:** push to `chr-5-7-phase0-run-locally` → Render auto-deploys (~2–4 min) at `cirqlback.onrender.com/cirql`; confirm the new `/assets/*.js` bundle hash.

Phase 2 — deferred / next

- **CHR-245** — retire the platform-wide arcade reward system → unify on sparqs (big blast radius; own pass, owner sign-off).
- **M8 remainder** — DM channel + full channel switcher + profanity filter + persistent report/block (CHR-248/249).
- **M11 wishlist** — emotes (CHR-260, roster ready), cutscenes/mini-journeys (CHR-264/262), Hearth décor + visiting (CHR-259), tapped-shops→landmarks (CHR-261), jump/hop (CHR-263).
- Real-world tap → sparqs bridge (the moat: a café tap widens the shared sea).